Unraveling the Mechanical Regulation of Lymphatic Button and Zipper Junctions: *In Silico* and *In Vitro* Insights

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Introduction

The lymphatic system is a specialized vessel network essential for fluid balance, immune defence, and macromolecule transport. While the cardiovascular system continuously circulates blood in a loop, the lymphatic system flows strictly in one direction, draining excess fluid from tissues back into the bloodstream. Lymphatic endothelial cells (LEC) within this network connect via two distinct cell-cell junctions: discontinuous "button-like" junctions at initial lymphatic capillaries which allow fluid and cells to enter easily, and continuous "zipper-like" junctions in downstream collecting vessels which prevent leakage as lymph flows toward systemic circulation [1]. In this work we combined an *in silico* and *in vitro* study to investigate the underlying mechanisms governing the mechanical behavior of these cell–cell junctions and how interactions with macrophages and cancer actively modulate junctional stability.

Materials and Methods

We developed both a standard 2D culture model using LEC monolayers and a 3D microfluidic device that recreates a tubular lymphatic vessel embedded within an extracellular matrix [2, 3]. In both systems, we evaluated key microenvironment variables by comparing control conditions, lung adenocarcinoma monocultures (A549), and co-cultures with M2 macrophages. These experimental models were integrated with advanced image analysis and computational simulation tools. In parallel, we developed a finite element model of isolated cell–cell junctions to investigate the mechanical behavior driving button and zipper junction dynamics.

Results and Discussion

We found that the tumor microenvironment actively regulates lymphatic junctions in 2D systems. Exposure to tumor cells alone or in co-culture with M2 macrophages induced junctional zippering, shifting the discontinuous "button-like" junctions, characteristic of initial lymphatic capillaries into continuous "zipper-like" junctions, typical of collecting vessels, alongside reduced VE-cadherin levels (Figure 1). Notably, in 3D microfabricated cylindrical LEC monolayers, no significant changes occurred in vascular permeability, vessel diameter, or tumor spheroid size across these conditions. Complementary finite element modeling (FEM) analysis revealed that button-like cell–cell junctions exhibit distinct mechanical behavior, with their functional responses varying according to applied stress levels.

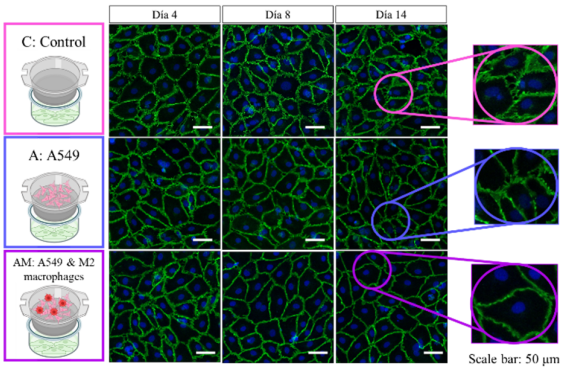


Figure 1: Representative fluorescence images of LEC monolayers after 4, 8, and 14 days of culture in (C) control medium, (A) A549 cancer cell culture, and (AM) A549 cancer cell and M2 macrophage co-culture. Green: VE-cadherin (Alexa Fluor 488). Blue: nuclei (DAPI). Scale bar 50µm.

Conclusions

Our findings reveal how tumor microenvironment factors regulate lymphatic endothelial junction mechanics. Validating this dual 2D/3D experimental framework alongside computational simulation tools provides a robust, versatile platform with great potential for future research into lymphatic mechanobiology and targeted cancer therapies.

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